

The Sound of Reference

Nirmalangshu Mukherji^{*,†}

ABSTRACT

Noam Chomsky and his colleagues have proposed that human languages are basically designed for construction of inner thoughts so that individual humans can plan ahead, thus giving selectional advantage to the species. The sound part of language in this view is ancillary, it is meant only for the externalization of thought. Call it a ‘thought-only’ view of language. We challenge this view by showing that the thought-only conception of language cannot accommodate the referential function of languages. We probe the issue with a focus on what philosophers call *Singular Terms* that enable a language-user to pick out, identify, and draw attention to an individual object. The presence of singular terms in a human language not only gives a shape to human thoughts about individuals, it ensures that such thoughts are shared between conspecifics. Since such sharing cannot be accomplished without explicit articulation, the sound of singular terms is essential for human language, it is not ‘ancillary.’

The standard view of human language is that it is a system of unbounded sound-meaning correlations. In this view, languages have two parts that correlate in systematic ways: a sound part, or more broadly, a sensorimotor part, and a meaning or thought part. To accomplish this effect without bound, the language system employs a generative or computational device, currently called *Merge* (Chomsky 1995), to construct structured expressions from a finite stock of lexical items, where each item consists of a cluster of syntactic, semantic, and phonetic features. The syntactic features essentially drive the language-internal computation. After each cycle of computation, the output is transferred to two language-external systems: the assembly of phonetic features is transferred to the sensorimotor (SM) systems and the assembly of semantic features is transferred to the conceptual-intentional (CI) systems.¹ In this paper, I will not be directly concerned either with the formal system itself or with its individual parts. I will be primarily concerned with a recent claim about what the system is for.

Noam Chomsky and his colleagues have claimed in recent years that human languages are primarily designed for construction of ‘inner thoughts’ (Chomsky 2007; Hauser 2009; Chomsky and McGilvray 2012; Chomsky 2015; Berwick and Chomsky 2016; Huybregts 2017; Huybregts 2020). These thoughts consist of cognitive images and conceptions of possible worlds which, according to these authors, do not require the sound part. Chomsky holds that these thoughts enable individuals to interpret their experiences, and make inferences from these

^{*}Former Professor, Department of Philosophy, University of Delhi, IN

[†]Corresponding author: Former Professor, Department of Philosophy, University of Delhi, IN.
Email: nirmalangshu.mukherji@gmail.com

interpretations, for planning and organization of action. Such abilities thus give selectional advantage to the species for its survival and eventual proliferation. For this primary purpose, the computational processing from the lexicon to the CI system is what the design of language is basically for. In this paper, we call this doctrine the *thought-only* view of language.

In contrast, the sound part of language, that is the computational processing from the phonological component of the lexicon to the SM interface, is claimed to be ‘ancillary’ in the sense that this component may be viewed as added to the basic language system for externalization of thoughts. Mark Hauser (2009) suggests that language evolved for internal thought and planning and only later was co-opted for communication. According to Hauser then the devices for externalisation were either absent or muted prior to the emergence of communication.²

Some authors have challenged this view by showing that human languages are typically learned from interactions between conspecifics which is impossible without the sound system (Studdert-Kennedy and Terrace 2017). Others argue that linguistic operations typically work on lexical concepts which need some language-specific phonological representations (Vicente et al. 2025; Vicente and Collins 2025). It seems to me that, even if these objections are plausible from what we currently know of languages, the objections may be deemed inadequate since they presuppose that the current state of human languages exhibit the very idea of language since its origin. There is no doubt that once the sound system was attached to the linguistic system, the total language system could have developed to adopt and adjust to the additional advantages (and disadvantages) offered by the sound systems. Chomsky cannot miss that obvious point.

Chomsky’s point rather is that a theoretically salient conception of language was available independently of the sound systems, perhaps even prior to the attachment of the sound system to the basic linguistic system. To that end, Chomsky in fact proposed that a thought-only language constituted of conceptual ‘atoms’ in a word-like format which were not really words because they did not have any phonological component either for lexicalizing concepts or for conspecific learning. Hence, the thought-only view of language remains unaffected unless it is shown that this view of language is difficult to conceptualize without the sound system. Hence a more foundational criticism of Chomsky’s view needs to target the very idea of a thought-only language. In this paper, I will argue that the thought-only view of language fails to incorporate the familiar referential function of singular terms.³

1. REFERENCE AND SINGULAR TERMS

It may be questioned whether the notion of reference is a relevant linguistic phenomenon for Chomsky’s conception of language at all. Chomsky has long maintained that human languages do not have a referential/denotational relation with anything either internal or external to the mind (Chomsky 2000). In this sense, according to Chomsky, human languages do not have semantics, they only have syntax and pragmatics.⁴ Therefore, it may be argued that we cannot complain about the failure to accommodate some notion of reference within the Chomskyan conception of language. Hence, at least a brief discussion on the notion of reference is warranted.

There is a distinction between two notions of reference in the functioning of language. For Chomsky, reference means some sort of a direct—perhaps causal—relation between a word and some extralinguistic object, typically something in the physical world or some nonlinguistic mental object; semantics is then viewed as the study of these direct word-object relations. Suppose we agree with Chomsky that human languages, unlike animal communication systems and some artificial symbol systems, do not have a direct (causal) relation between a sign and some physical object.⁵ For example, although the word *London* mentions a city located in a definite physical space, there is no referential relation in the Chomskyan sense between the word *London* and that physical space; we may as well say that London was rebuilt across the river (Chomsky 2000).

However, if we think of the notion of reference somewhat broadly to indicate our ability to think and talk about the world, reference in this sense must be taking place in most cases of language-use even if nothing in the world answers to the reference-indicating expressions so used.⁶ I wish to focus on the ubiquitous presence in every human language of what philosophers call *singular terms* such as proper names, singular definite descriptions, and singular demonstratives. The referential function I have in mind is the ability of a competent speaker to use these terms to *pick out* and *draw attention* to individual objects.⁷

When a singular term is used in an expression, the term signals the thought of an individual; in that sense, the term is *about* an individual; we may say that the term *picks out* an individual.⁸ However, as we will see, singular terms not only generate thoughts about some individuals, their use typically involves the additional act of *drawing attention* of a conspecific to these objects. Uses of a singular term thus conceal at least two functions: generating thought of an individual and directing attention to the individual. We may think of the composite function as a referential function in the second sense.

There is a strong intuition that the notion of reference in this second sense is indeed available somehow even if humans use language primarily for construction of inner thoughts of possible worlds, cognitive images, and the like. This is because the (evolutionary) significance of these inner thoughts largely accrues from their putative connections with the world. Use of language will not have survival value if the human mind created mostly fictions or nonsense. It is in the purported real connections with the world that our plans and goals have a chance. So the intuition is that there are likely to be instruments in language itself that are more directly geared to reflect these (putative) connections. As epistemic agents, humans are thus armed with singular terms so that the extralinguistic knowledge of an object may be shared and, in the favourable cases, relied upon.

Reference then is a function of some vehicle(s) of reference where these vehicles are parts of a symbol system. Intuitively, just the ability to locate or act on a particular object does not amount to referential ability. Even though a tiger searches for a definite shelter or a particular mate in the world, it is counterintuitive to say that the tiger ‘refers’ to the shelter or the mate because the tiger is not a symbol-manipulator (Hinzen 2016).⁹ In the case of singular reference in natural languages, these vehicles take the shape of proper names, singular definite descriptions, and singular demonstratives, as noted.

Incidentally, Chomsky appears to reject the idea of a linguistic vehicle for reference because Chomsky holds that words don’t refer, people do; just as legs don’t walk, people do (Chomsky 2000). We may note that, in claiming this, Chomsky indeed upholds some notion of reference as action, even if he denies the idea of reference as a property of words. We will see that Chomsky’s notion of reference is sufficient for grasping the referential functioning of singular terms. For now, there seems to be no conflict between the notion of reference as action and the idea of a linguistic vehicle for reference. People need to use words to refer, just as people use legs to walk. If there are no legs, people can’t walk. On similar grounds, there must be some device(s) for achieving reference in a language. According to this perspective on reference, creatures can refer if their thoughts about the world have word-like constituents acting as vehicles of reference. In the current state of natural languages, singular terms are among these vehicles. I will suggest that, insofar as singular terms contribute to singular reference, these resources are absent in a thought-only language. To emphasize, this paper concerns the phenomenon of the use of singular terms; the notion of singular reference is used to label the phenomenon.

2. THOUGHT-ONLY LANGUAGE

So the issue is: what resources for reference are available to a silent creature that is endowed just with a thought-only language? We need to ponder on the issue because, *prima facie*, there

is no necessary connection between having sound and achieving reference; hence, we cannot reject the thought-only idea of reference by fiat just because of our unfamiliarity with a soundless language.

Here we face a problem. Our common understanding of referential devices consists of word(s) which are sound-meaning correlations; hence, it is unclear what it means to be endowed with a natural language which does not have a sound system (Mukherji 2021). Indeed, it is difficult to make sense of Hauser's observation that language evolved initially only for thought and planning. The difficulty arises because we do not really know which resources were available in the early stages of language such that they were adequate for structuring thought that guided planning etc. In particular, we do not know what constituted the features of the early lexicon and how they were marked for the computational system to work on them.¹⁰

However, the empirical issue of how human languages were organized in the earliest stages of evolution of language need not detain the inquiry here. All we want is some conception of a language, or a language-like device, that is primarily designed for thought and planning. To sidestep the empirical problem, then, we may conduct a thought experiment by introducing artificial referential devices. Just as people are able to walk with artificial legs, we may think of an artificial referential device which satisfies the conditions of a thought-only language, and see how it works.

To that end, we may borrow some elements from another thought experiment proposed by Tanya Reinhardt (2006). Reinhardt develops Chomsky's narrative about the origin of language. According to Chomsky as noted, we may imagine a pre-human hominid with the SM and CI systems, very similar to ours, already in place. Language emerged when the generative device Merge was inserted in the hominid's brain.¹¹ Reinhardt elaborates on the SM and CI systems available to the hominid prior to the emergence of language. According to Reinhardt, the hominid's pre-linguistic system contained the following: a system of concepts similar to that of humans, a sensorimotor system for perceiving and coding information in sounds, an innate system of logic, an abstract formal system, inventory of abstract symbols, and connectives, functions, and definitions necessary for language.

It may be questioned whether such rich logical resources were available to the pre-linguistic hominid in the absence of Merge (Mukherji 2010; 2022). However, the objection does not apply to the linguistic creature, albeit soundless, under study here. So, let us take away the sensorimotor system from the pre-linguistic hominid and add the generative operation Merge; we can now ascribe the rest of Reinhardt's resources to the resulting hominid to reach our desired silent linguistic creature. Call her *Proto*.

The items in Reinhardt's list for CI systems look very much like the resources in the familiar Frege-Russell first-order logic.¹² Suppose we adopt the notation of first-order logic for describing Proto's linguistic organisation. It will be methodologically convenient to have a reassuringly familiar symbol system in hand for describing Proto's thoughts, especially in view of Chomsky's claim that Proto's CI system was very much like ours. Furthermore, even if we assume that Proto's language is unspeakable, we need to mark its symbolic objects with some articulable signs so that we can talk or write about Proto's thoughts. Therefore, in what follows, whenever we mention an expression of the thought-only language, we assume that these are denizens of the dark. The signs we thus use may be viewed as current theoretical devices for talking about these thoughts. The notation of first-order logic may be viewed as such a theoretical device while staying close to Proto's possible mental design.

Since Proto was endowed with an advanced CI system that may enter into thought-only computation, we are in a position to ask: what resources of reference did Proto have? As we saw, the referential function of a language is routed through thoughts about individual objects. In other words, once we are assured that Proto is able to entertain singular thoughts, we may

proceed to ask if such thoughts are adequate for singular reference. Since Proto was endowed with language, she was likely to have had singular thoughts as part of her inner thoughts. So, how are Proto's singular thoughts represented with the suggested logical resources?

Suppose we have the thought

- (1) ⟨Aha, ripe banana⟩

which may be represented in English with the usual Frege-quotes to highlight that it is an element of inner thought, not an articulated thought. I don't know how else to write down an inner thought in a natural language. As noted, it is likely that Proto was endowed with at least the resources of first-order logic. So, it is plausible that our ripe-banana thought may be captured in something like

- (2) $(\exists x)(Bx \ \& \ Rx)$

which is a singular thought about a ripe banana.

Similarly, within the same logical resources, Proto may have had adequate representation of more advanced thoughts such as

- (3) ⟨Exactly the mate I want⟩.

In the ascribed symbolism, the English thought may be represented as

- (4) $(\exists x)(Mx \ \& \ \forall y(My \rightarrow x=y) \ \& \ Wx)$

which, as Bertrand Russell (1905) told us, is equivalent to a definite singular thought. While we are at it, suppose we endow Proto with resources a bit beyond classical first-order logic. Thus suppose the Russellian resources are enriched with suggestions from Richard Montague for Proto to entertain the thought

- (5) $\lambda F[\lambda G \ \exists y[\forall x[Fx \leftrightarrow x=y] \ \& \ Gy]]$,

where 'F' and 'G' are predicate-variables (Dowty, Wall, and Peters 1981). Suppose they take the predicates *mate* and *want* as values. This would be a genuine singular thought expressed in a term,

- (6) ⟨My uniquely wanted mate⟩.

The preceding logical constructions ascribed to Proto suggest that a variety of familiar singular thoughts are available to it within the resources of a thought-only language. Arguably, that is all the linguistic resources we have for achieving singular reference *in* language, the rest of the connections with the world are to be achieved with nonlinguistic resources, such as perceptual information. The argument in fact applies not only to the artificial thought-only language under study here, it also applies to the thought-part of natural languages such as English. I guess that is Chomsky's point. In this picture, the working of language towards singular reference ends with thought-only constructions; the rest of the directedness towards the world—the *aboutness* of linguistic expressions—is achieved by plugging the thought-only expression to external systems.¹³ With a thought-only language in hand, we are in a position to inquire how singular terms fit into this scheme.

The striking thing about Proto's resources described so far is that Proto had sophisticated singular thoughts in mind purely in general terms because its resources do not explicitly contain what are known as 'singular terms' in the philosophical literature. As noted, by 'singular terms' I mean symbolic devices such as names like *Socrates*, (singular) definite descriptions like *the ancient Greek philosopher*, and (singular) demonstratives like *that philosopher*. For the restricted purposes of this paper, I am setting aside all other devices in language, such as pronouns, indexicals, etc., which help a language-user to identify and mention individual objects and events. Given Proto's ability to represent inner singular thoughts with the suggested logical resources alone, the question arises as to whether singular terms in the stated philosophical sense are available in its language.

Indeed, with very much like the resources of first-order logic in hand, Willard Quine (1960) once proposed that singular terms are not needed in a 'regimented' language; as Quine phrased it, singular terms may be 'eliminated' from a 'scientifically-adequate language.' Therefore, since the suggested Reinhardt resources apparently constitute a regimented language in Quine's sense, these resources are enough for Proto to entertain singular thoughts in sufficient variety; hence, classical singular terms are apparently not needed in Proto's regimented language as well. Setting aside the issue of what kind of linguistic terms are needed in science, if any, it is interesting to study Quine's method of eliminating singular terms in a regimented language.

Cutting through a wide range of empirical and technical details, Quine's basic idea is to adopt Russell's Theory of Definite Descriptions. As mentioned, Russell (1905) suggested that a sentence of the form

(7) The F is G,

with a singular definite description in the subject position, may be recast in the notation of first-order logic as

(8) $(\exists x)(Fx \ \& \ \forall y(Fy \rightarrow x=y) \ \& \ Gx)$,

meaning roughly that there is exactly one F and it is G. The original definite description, *The F*, no longer occurs as a unit in the Russellian reconstruction. According to Quine, definite descriptions as singular *terms* thus stand eliminated from the Russellian regimented language.

Next, take a name such as *Socrates* and construct a predicate *is socrates* (or *socratises*) to yield the structure

(9) The object *x* such that *x* is socrates (or, *x* socratises).

Notice that *socrates* with small *s* shows that the original name is now treated as a general term. As Strawson (1961) pointed out, the only singular term in Quine's canonical form is the (individual) variable. Now the resulting sentence with general terms will submit to Russell's analysis in obvious ways. With singular demonstratives, the task is straightforward: *This/that man* is to be rewritten as *The man here/there*, which can then be subjected to Russell's analysis.

If Quine's analysis of singular terms is right, we need not bother if these terms are available in the thought-only language under study insofar as the language is sufficient for entertaining singular thoughts. Recall that we viewed the referential function as a composite function achieving two things, generating singular thoughts and drawing attention to individual objects. We can now say that Proto's thought-only language achieves the first function of generating singular thoughts. We will now examine what it takes to achieve the composite function.

3. STRAWSON'S DISTINCTION

Are human languages Quine-regimented such that the mentioned singular terms are essentially redundant and that they may be replaced by expressions consisting entirely of general terms? As we saw, the key to Quine's strategy of elimination of singular terms is to paraphrase a singular definite description of the form *The F* into a first-order canonical form via Russell's Theory of Descriptions. Russell viewed a sentence with a singular definite description in the subject position, *The F is G*, as a quantified expression with an existential clause. In particular, *The F* is viewed as a quantified expression meaning one and only one *F* or exactly one *F*; David Kaplan (1972) called it Russell's 'fundamental equivalence' (Mukherji 2010, 3.5.3). In what follows, I will be concerned only with this step of Russell's theory.¹⁴

Following the work of Peter Strawson, I will suggest that the linguistic functions of *The F* and *Exactly one F* do not match. Let me clarify the textual basis for this view. In his classic paper, Strawson (1950) opposed Russell's theory by arguing that Russell's fundamental equivalence cannot be viewed as a paraphrase of the original definite description. Rather, the expression *one and only one F*, or its Russellian canonical rendition, may at best be viewed as a presupposition or condition for asserting a sentence with a definite description *The F is G*. In my view, in a later paper, Strawson (1961) clarified the idea of presupposition somewhat differently to grasp the distinction between *The F* and *one and only one F*.

In his 1961, Strawson suggested that expressions like *one and only one F* signal new resources, while expressions like *The F* draw on established resources (403). The expression *resources* signals the knowledge available to language-users for identifying objects or referents in the world. According to Strawson, there are two different ways in which referents are picked out or identified by linguistic means. In the first way, a speaker uses a linguistic expression to identify a referent that the speaker has in mind and introduces the referent for the hearer. In this case, the speaker's introduction of the referent is the only access for the hearer to identify the referent that the speaker wishes to talk about; as such, the content of the linguistic expression used by the speaker supplies a *new* resource for the hearer.¹⁵ Hence, if the linguistic expression remains unarticulated, the intended referent will not be available to anyone beyond the speaker.

In the second way, the speaker uses an expression to draw the attention of the hearer to a referent which, according to the speaker, has been already identified by the hearer by independent means. The identification of the referent is thus presupposed in the resulting discourse.¹⁶ According to Strawson, since the target referent is already established prior to the use of the concerned expression, the use forestalls the question 'which one' for the hearer. Although Strawson made the distinction primarily for singular definite descriptions, we will see that the distinction applies to other singular terms as well.¹⁷

Let me try to understand Strawson's distinction with straightforward linguistic intuitions. Let us suppose a school-teacher asking in a history class: 'how many kings of France died at the guillotine?' Supposing just one king of France died at the guillotine, an appropriate and correct response would be, 'one', the answer signals a quantity. If the teacher pursues, 'isn't that more than one?', an appropriate answer would be, 'no, exactly (or just or only) one.' The point is, the question requested a number, possibly a unique one, and that request is not fulfilled by uttering *the king of France*. Suppose now the teacher asks in a different scenario: 'which one of them ruled from Versailles?' Now an appropriate and correct answer would be, *the king of France*. In this case, it would be totally inappropriate to respond with *one and only one king of France*.

So the basic distinction between quantified phrases like *one and only one king of France* and definite descriptions like *the king of France* is that, while the former responds to a 'how many' question, the latter is a response to a 'which one' question. This difference in the function of the concerned expressions leads to Strawson's distinction between introducing resources and

drawing upon independently established resources. Thus, in the first scenario, the teacher was testing if the children can introduce the reference properly; in the second scenario, the teacher was testing if the children can identify the referent by drawing on their established resources.

The contrast between quantified expressions such as *exactly one* and definite descriptions is also illustrated in the so-called anaphoric uses of definite descriptions in a discourse (Heim 1982). The quantified phrase *a man* in the sentence

(10) A man came to see me

establishes a new resource, arrival of a man. We may then use the established resource to say,

(11) The man was drunk,

mentioning the man already identified. This form of dependent use is not restricted to definite descriptions in the subject position. The request

(12) Please give me the book

is used correctly when the hearer is able to identify a book from independent resources; otherwise, the hearer is likely to ask, 'which one?'

Let us suppose that singular definite descriptions are constituents of inner singular thought that are aided by perceptual systems and other resources of knowledge for identifying referents in the world; that happens with any purported talk about the world anyway including the talk with expressions of the form *one and only one F* and *exactly one F*. However, according to Strawson, uses of definite descriptions *also* typically involve drawing the attention of the hearer to objects whose identification has been established on independent grounds.¹⁸ Thus, both the uses of *exactly one F* and *the F* represent some object that the speaker has in the mind.

The crucial difference between them is that, in the use of *exactly one F*, the singular thought is there in the mind of the speaker irrespective of whether there is a hearer or not. In contrast, definite descriptions are 'name-like' (Strawson 1995) in that a speaker using *The F* assumes that, when directed by the speaker's use of the description, the hearer can locate the referent mentioned by the definite description from his own resources of knowledge. In that sense, the use of a definite description requires a speaker-hearer context and, for that reason, it is unavailable in a thought-only language. On this point, Strawson's distinction differs sharply from two other apparently similar distinctions concerning meaning and function of singular terms.

Consider briefly the distinction between two uses of definite descriptions proposed by Keith Donnellan (1966). Donnellan distinguished between what he called the *attributive* and the *referential* uses of definite descriptions: in an attributive use, some individual, whoever she or it is, is pointed out in a general way; in a referential use, a particular individual is mentioned. For example, the sentence

(13) The murderer of Smith is insane,

may be used either attributively or referentially depending on the background stories that accompany such uses. It can be argued that Donnellan's distinction, including referential uses, possibly applies to most quantificational structures in a language, such as *a man* and *every woman*. Apart from quantified expressions, Donnellan's distinction perhaps applies even to some names and some demonstratives with a certain twist (Mukherji 1996).

Most importantly, nothing prevents Donnellan's distinction, including referential uses, to apply to singular thoughts in a thought-only language insofar as a singular thought mentions an individual in a general way or picks out particular individuals via nonlinguistic resources. In contrast, Strawson's idea of drawing on independent resources applies only to definite descriptions and other singular terms, but they do not apply to quantified structures such as *exactly one F*. As a result, as already indicated, Strawson's idea does not apply to thought-only languages.

Strawson's distinction also differs on this point from recent challenges to the Fregean conception of meaning of some of the singular terms. Cutting through many details and much controversy, many authors claim that ordinary proper names do not have Fregean sense at all; if anything, they only have reference such that the referent, if any, of a proper name directly contributes to the truth-condition of a sentence containing it without routing the expression via Fregean sense (Kripke 1980; Recanati 1993); such a view is sometimes called *Direct Reference*.

Although complex demonstratives like *that F* do contain a Fregean conceptual component *F*, David Kaplan (1989) suggests that in a typical use of a demonstrative associated with a demonstration, the common noun in a complex demonstrative may be 'whispered aside'. So, complex demonstratives also fall under direct reference.¹⁹ Since it is commonly believed that definite descriptions like *the F* contain a descriptive 'blue-print' for achieving reference (Almog 1980), there is supposed to be a sharp distinction between the Fregean (conceptual) character of definite descriptions, and directly referential character of proper names and demonstratives.

In the next section, I will briefly sketch a very different way of distinguishing between definite descriptions, demonstrative and proper names in terms of their referential functions. For now, the important feature of the Fregean and non-Fregean views is that they are both compatible with a thought-only conception of language; these views differ only in what a singular term contributes to the compositional meaning of a sentence. In that sense, they are just different views of what constitutes the content of Proto's thought. So both Donnellan's distinction between uses of singular terms, and the distinction between Fregean and non-Fregean views of them, are compatible with thought-only language.

4. THE SOUND OF SINGULAR TERMS

Returning to Strawson, the example of definite anaphora, apart from illustrating Strawson's distinction, also suggests that the distinction cannot be formally understood in terms of the grammatical category of determiners. Both *a* and *the*, and a host of other linguistic items are classified as determiners in grammar, but, in terms of Strawson's distinction, *a* introduces new resources while *the* draws upon resources already introduced. To highlight this distinction, let us call general determiner structures *det-structures*, and structures with definite descriptions *def-structures*. Thus, although *def-structures* are *det-structures* in form, *def-structures* are a proper subset of *det-structures* in function. As we will now see, all singular terms as specified above may be viewed as expressions with *def-structures* functioning as referential devices that draw on already established resources.

Definite descriptions of course are *def-structures* by definition. Demonstratives like *that* are also *def-structures* in a similar way. Like definite descriptions, complex demonstratives also have the form of a determiner followed by a common noun, *that F*, and they are typically used when the hearer can pick out a referent independently, usually aided by a pointing gesture. In that sense, demonstratives indicate an object or event independently identified. Thus, *The F* and *That F* share this general referential function of drawing on established referents. However, this generality does not preclude subtle differences in application between *the* and *that*. For example, we can say,

(14) Give me that book, that book and that book,

but we cannot say,

(15) *Give me the book, the book and the book.

Further, although we can say

(16) give me that,

we cannot say

(17) *give me the.

Although both *that* and *the* appear in def-structures, these subtle variations suggest that their uses as referential expressions are fine-tuned to the variations in resources of knowledge for efficient functioning of language.²⁰

Interestingly, it appears that proper names may also be viewed as def-expressions. The most immediate case of proper names as def-expressions are definite descriptions which have 'grown' into names, as Strawson put it:

(18) The White House, The Holy Book, The Himalayas.

Even regular names can form an overt def-structure with *the* or *that*:

(19) The Noam Chomsky.

The phenomenon occurs across languages. In Bangla, we can say,

(20) Rām-tā elo-nā

Ram-the/that come-not

That Ram didn't come/show up,

where *Rām* is a name and *-tā* is a postpositional definite item.

Such examples suggest that even names without overt def-structures may sometimes be viewed as covert definite items with an overt null def component, *insofar* as their noted referential function is concerned.²¹ Under this conception of referential function, we may view most names, perhaps all, as def-structures because we can think of circumstances in which a common name like *John Smith* is prefixed with *the*. We may thus conjecture that, at the output of grammar, singular terms are just (subliminal) def-structures.

In pragmatics, they fan out as overt definite descriptions, proper names, and demonstratives with many variations in between. They all serve the same primary function of drawing on independently established resources. But individually, they cover different segments of the spectrum of resources so that the speaker is able to draw attention of the hearer to specific areas of the spectrum. The invocation of a spectrum of resources suggests a very different perspective on how singular terms function in languages.

For instance, once the use of definite descriptions is delinked from Russell's quantificational picture, it is questionable whether definite descriptions may be included in the Fregean class

of expressions with conceptual content (sense) at all. More specifically, the use of *the* does not signal some quantity bearing the sense of the expression, it is something of a pointer to an existing resource. No doubt, the pointing is assisted by the rest of the phrase consisting of *F* to narrow down the general direction of search, as with occurrences of *F* in complex demonstratives. But then, more or less on par with demonstratives, much of the *F*-part in a definite description as well may be ‘whispered aside’ for an ‘indexical’ phrase such as *the man over there*. For nonindexical descriptions, how much conceptual content goes into the *F*-part depends on the speaker’s estimate of the hearer’s resources. Sometimes *the man* will do, on par with *that guy*; in other circumstances we need to say *the man I was talking about yesterday*. So, there is at best a difference in the grade of ‘blue-print’ between standard definite descriptions and complex demonstratives. From this perspective, definite descriptions, demonstratives, and proper names form a graded distribution within the spectrum of def-structures. A detailed study of the phenomenon is beyond the scope of this paper.

For this paper, the most compelling aspect of the phenomenon of singular terms is that they are directed at hearers. So, they are a part of the conversation between conspecifics. Hence they require the sensorimotor apparatus in the absence of telepathy. Proper names are typically ascribed to objects we already identify; therefore, it is hard to see what the name *itself* contributes to the content of our singular thoughts in either Fregean terms or as direct reference. As David Kaplan (1973) observed, there is really no need of proper names in a language except for calling kids to dinner.

Demonstratives are typically used to indicate objects we already see; so we use them for drawing attention of the hearer to something we assume is visible to the hearer as well; hence, they are typically accompanied by a pointing gesture.²² As a result, in many cases, the common noun maybe ‘whispered aside’ or not used at all, as noted (Kaplan 1989). As emphasized, definite descriptions are used when the speaker is relatively certain that the hearer can pick out the referents from the indicated general area from her own perceptual, contextual and other resources of knowledge (Strawson 1961).

The preceding analysis explains why Proto, with a thought-only language, did not possess singular terms in her linguistic repertoire simply because she did not need them in the absence of language-speaking conspecifics. When we use a definite description, demonstrative, or a name to direct the attention of the hearer, we already have certain objects in our minds to which we direct attention of others. Granting that Proto had the Reinhart-resources to frame her singular thoughts, there is no additional need for her to be endowed with singular terms. Proto plans her moves in the world alone and in silence. Proto’s solitariness can be traced to the restrictions of her Proto-language. Lacking a sound system, Proto was not endowed with a human language, not yet.

5. CONCLUDING REMARKS

I will conclude this discussion with two quick observations on two largely open questions on the character of human languages.

First, we ascribed singular thoughts to Proto by virtue of her Reinhardt-resources; you can ascribe whatever content you want to a bound variable. Probably due to the pull of tradition, we also ascribed singular thoughts to humans equipped with singular terms. However, by now, it is unclear what a singular term contributes to the compositional meaning of a singular thought. At best, the term indicates some directions for interpersonal use in contexts of shared knowledge. It appears that the classical notion of a singular thought, with either Fregean or Russellian content, has no place in human languages. Which notion of singular thought, if any, is available in human languages, thus remains an open question.

Second, the design of human languages that begins to emerge from the preceding observation seems to be very close to a conception of human languages that Chomsky has long advocated, but has never fully discussed to my knowledge. As we saw, Chomsky holds that the words of a language do not refer; people engage in the act of reference by using words in appropriate contexts (Chomsky 2000, 132). In this sense, human languages have only syntax and pragmatics as parts of their design (Chomsky 1995, 26). From this general perspective, Chomsky suggests that human languages are instruments for internal computations that encode instructions for use in an actual speech community (Chomsky 1992, 223). From what we saw above regarding singular terms, a Strawsonian treatment of them appears to endorse Chomsky's perspective. Whether the perspective can be generalised for the entirety of human languages also remains an open question.

NOTES

1. Among the sensorimotor systems, I will be concerned only with the sound system in this paper.
2. Riny Huybregts (2017; 2020) claims that San populations in Africa acquired click phonemics only after they separated from the original Khoisan group. An evaluation of this claim is beyond the scope of this paper.
3. Elsewhere (Mukherji, in preparation), I argue that the sound system is essential for formation of concepts; concepts enter human thought *as* lexical concepts. So, this is also a direct objection to the very idea of a thought-only language. However, I wish to examine a different objection in this paper.
4. I am setting aside the recent claim that some grammatical phenomena, such as certain phasal movements, are geared to the phenomenon of reference (Sheehan and Hinzen 2011; Hinzen 2016); in this view, reference is a phenomenon built into the functioning of the grammatical system itself. It is unclear if these otherwise most interesting claims fall under Chomsky's conception of reference.
5. As Chomsky (pc) puts it, all that happens in the animal case is that "rustling of the leaves elicits a 'warning cry', hormonal changes elicit some vocal noise."
6. Fictional use of expressions is a case in point; *Sherlock Holmes* is certainly reference-indicating in this sense. As Peter Strawson pointed out, even if nothing answered to *Sherlock Holmes* in history, the name did pick out an individual in Conan Doyle's story (*his* story). But to say this is not to deny that human thought sometimes fails to achieve reference and, in those rare cases, it is not about anything at all: *The colourless green idea sleeps furiously*.
7. Chomsky holds that the output of the language system interfaces with the Conceptual-Intentional (CI) systems. My concerns in this paper are perhaps more directly related to the intentional systems.
8. I will use the expressions *thoughts about individuals* and *singular thoughts* interchangeably without meaning that these individuals themselves are constituents of these thoughts. Thus, I am not concerned with 'singular thoughts' mentioned in the literature on direct reference (Salmon 1986; Kaplan 1989; Recanati 1993). In general, I will not discuss the cognitive and epistemic conditions associated with the formation of thoughts about individuals; see Openshaw (2022) for a recent discussion.
9. Therefore, it is unclear which nonhuman communication systems Chomsky has in mind in claiming that nonhuman systems may be endowed with the first (Chomsky's) sense of reference.
10. Moro (2016, ch. 8) makes the interesting but not entirely surprising claim that a 'sound envelope' accompanies the process of thinking even when someone is reading something

mentally, in silence. Once words are learned as units of sound-meaning correlations, inner images of sound are likely to accompany words even in writing, much like the reading of a musical score.

11. In this paper, I will not be concerned with the empirical dimensions of Chomsky's evolutionary picture (Mukherji 2022).
12. It may be questioned whether a creature so early in symbolic evolution may be ascribed such rich CI resources. John Bolender (2007) traced the cultural explosion in the Upper Palaeolithic era, at about 40 KY, to the human ability to use something like a system of Russellian quantified logic. By that time, human language was already firmly in place for several tens of thousands of years.
13. I am assuming, essentially out of ignorance, that thought-only constructions contain the resources for effecting the required plug-ins.
14. Even after a century of discussion on Russell's ground-breaking paper on denoting phrases, controversies about Russell's theory continue. I will generally stay away from these controversies; but see Mukherji (1989; 1995; 2010).
15. Following Heim (1982), Diesing (1992), and related work, it is commonplace in the pragmatics literature to characterize indefinite and definite descriptions in terms of a new/familiar distinction which sounds similar to Strawson's distinction. However, to my knowledge, the literature fails to stress that the distinction typically applies to the resources available to the *hearer* in an interpersonal context.
16. The notion of presupposition concerns prior knowledge. It is totally unclear if such knowledge must take the Russellian form *one and only one*, etc. In my view, Strawson (1961), unlike Strawson (1950), advocates total separation of content and function between *The F* and its alleged Russellian canonical form; see Mukherji (1995) and Mukherji (2010, especially 3.5.3).
17. Arguably, 'Russellian' descriptions, such as *the least prime*, *the present king of France*, etc. appear to escape Strawson's view of descriptions. In my view, even these can be accommodated within the Strawsonian scheme, but I will not insist on the point here (Mukherji 1995).
18. It is unclear from the text if Strawson wants definite descriptions to have only this use of drawing attention to referents established on independent grounds. I am assuming that a speaker can draw the attention of a hearer only if the speaker has formed a singular thought with these descriptions in the first place without suggesting of course that these descriptions have a Russellian form. Hence I am suggesting that drawing the attention of the hearer is an additional aspect of the same use.
19. See King (1999) for a very different view.
20. Interestingly, on par with *give me that*, we can also say, *give me all*, *give me one*, *give me some*, *give me many*, etc. However, the expression *give me that* typically ensues in response to a 'which one' question, while the others are answers to 'how many' questions. So the commonality of use does not defeat Strawson's distinction. A detailed discussion of these issues is beyond the scope of this paper.
21. Longobardi (1994) suggests a marked difference between proper names and definite descriptions with respect to how they occur in a syntactic tree. This leads him to a Russellian view of semantics of proper names and definite descriptions. It is unclear if this view extends to Strawsonian referential function. The topic is beyond the scope of this paper.
22. There is some evidence (Povinelli 2000) that such pointing gestures towards objects in the visual field are apparently unique to humans.

REFERENCES

- Almog, J. 1980. "Naming Without Necessity," *Journal of Philosophy* 83(4): 210–42.
- Berwick, R. and N. Chomsky 2016. *Why Only Us: Language and Evolution*, Cambridge, MA: MIT Press.
- Bolender, J. 2007. Prehistoric Cognition by Description: A Russellian Approach to the Upper Palaeolithic, *Biol Philos* 22: 383–99. <https://doi.org/10.1007/s10539-006-9058-2>
- Chomsky, N. 1992. "Explaining Language Use," *Philosophical Topics* 20: 205–31. Reprinted in Chomsky (2000).
- Chomsky, N. 1995. *The Minimalist Program*, Cambridge, MA: MIT Press.
- Chomsky, N. 2000. *New Horizons in the Study of Language and Mind*, Cambridge: Cambridge University Press.
- Chomsky, N. 2007. "Approaching UG from Below," in U. Sauerland and H. Gärtner, eds., *Interfaces + Recursion = Language? Chomsky's Minimalism and the View from Syntax-Semantics*, De Gruyter Mouton. <https://doi.org/10.1515/9783110207552>
- Chomsky, N. 2012. *The Science of Language: Interviews with James McGilvray*, Cambridge: Cambridge University Press.
- Chomsky, N. 2015. *What Kind of Creatures Are We?* New York: Columbia University Press.
- Diesing, M. 1992. *Indefinites*, Cambridge, MA: MIT Press.
- Donnellan, K. 1966. "Reference and Definite Descriptions," *Philosophical Review* 75: 281–304.
- Dowty, D., R. Wall, and S. Peters, eds. 1981. *Introduction to Montague Semantics*, Dordrecht: D. Reidel.
- Hauser, M. 2009. "The Illusion of Biological Variation," in M. Piattelli-Palmarini, J. Uriagereka, and P. Salaburu, eds., *Of Minds and Language: A Dialogue with Noam Chomsky in the Basque Country*, Oxford: Oxford University Press.
- Heim, I. 1982. *The Semantics of Definite and Indefinite Noun Phrases*. Ph.D. thesis, reproduced by Graduate Linguistics Student Association, University of Massachusetts, Amherst.
- Hinzen, W. 2016. "On the Grammar of Referential Dependence," *Studies in Logic, Grammar and Rhetoric* 46(1): 11–33. <https://doi.org/10.1515/slgr-2016-0031>
- Huybregts, R. 2017. "Phonemic Clicks and the Mapping Asymmetry: How Language Emerged and Speech Developed," *Neuroscience and Biobehavioral Reviews* 81, Part B, 279–94.
- Huybregts, R. 2020. "Biting into Evolution of Language," *Journal of Language Evolution* 5(2): 175–83. <https://doi.org/10.1093/jole/lzaa003>
- Kaplan, D. 1972. "What Is Russell's Theory of Descriptions?" in D. Pears, ed., *Bertrand Russell: A Collection of Critical Essays*, New York: Doubleday Anchor, 227–44.
- Kaplan, D. 1973. "Bob and Carol and Ted and Alice," in J. Hintikka, ed., *Approaches to Natural Language*, Dordrecht: D. Reidel Publishing, 490–518.
- Kaplan, D. 1989. "Demonstratives," in J. Almog, J. Perry, and H. Wettstein, eds., *Themes From Kaplan*, New York: Oxford University Press, 481–564.
- King, J.C. 1999. "Are Complex 'That' Phrases Devices of Direct Reference?" *Nous* 33(2): 155–82.
- Kripke, S. 1972/1980. *Naming and Necessity, Three Lectures at Princeton University*. Reprinted Cambridge, MA: Harvard University Press, 1980.
- Longobardi, G. 1994. "Reference and Proper Names: A Theory of N-Movement in Syntax and Logical Form," *Linguistic Inquiry* 25(4): 609–65.
- Moro, A. 2016. *Impossible Languages*, Cambridge, MA: MIT Press.
- Mukherji, N. 1989. "Descriptions and Group Reference," *Journal of Indian Council of Philosophical Research* 6(3): 89–107.
- Mukherji, N. 1995. "Identification," in P. Sen and R. Verma, eds. *Philosophy of P.F. Strawson*, New Delhi: Allied Publishers, 32–55.
- Mukherji, N. 1996. "Loads on Reference," in A. Chakravary, ed., *Epistemology, Meaning and Metaphysics After Matilal*, *Studies in Humanities and Social Science* 3(2): 57–80.
- Mukherji, N. 2010. *The Primacy of Grammar*, Cambridge, MA: MIT Press.
- Mukherji, N. 2021. "On Sound-Meaning Correlation," in P. Kosta and K. Schlund, eds., *Keynotes from the International Conference on Explanation and Prediction in Linguistics (CEP): Formalist and Functionalist Approaches*, Peter Lang.
- Mukherji, N. 2022. *The Human Mind Through the Lens of Language: Generative Explorations*, London: Bloomsbury Academic.
- Openshaw, J. 2022. "Does Singular Thought Have an Epistemic Essence?" *Inquiry*, 1–24. <https://doi.org/10.1080/0020174X.2022.2155871>

- Povinelli, D. 2000. *Folk Physics for Apes: The Chimpanzee's Theory of How the World Works*, Oxford: Oxford University Press.
- Quine, W. 1960. *Word and Object*, Cambridge, MA: MIT Press.
- Recanati, F. 1993. *Direct Reference: From Language to Thought*, Oxford: Blackwell.
- Reinhart, T. 2006. *Interface Strategies: Optimal and Costly Computations*, Cambridge, MA: MIT Press.
- Russell, B. 1905. "On Denoting," *Mind* 14: 479–93.
- Salmon, N. 1986. *Frege's Puzzle*, Cambridge, MA: MIT Press.
- Sheehan, M. and W. Hinzen 2011. "Moving Towards the Edge," *Linguistic Analysis* 37(3-4): 1–54.
- Strawson, P. 1950. "On Referring," *Mind*, July. Reprinted in *Logico-linguistic Papers*, London: Methuen, 1971, 1–27.
- Strawson, P. 1961. "Singular Terms and Predication," *Journal of Philosophy*, 58. Reprinted in *Logico-linguistic Papers*, London: Methuen, 1971, 53–74.
- Strawson, P. 1995. "Replies to Mukherji," in P. Sen and R. Verma, ed., *Philosophy of P.F. Strawson*, New Delhi: Allied Publishers, 32–55.
- Studdert-Kennedy, M. and H. Terrace 2017. "In the beginning: A Review of Robert C. Berwick and Noam Chomsky's *Why Only Us*," *Journal of Language Evolution*, 114–25. <https://doi.org/10.1093/jole/lzx005>
- Vicente, A., J. Collins, and M. Jary 2025. "Thinking with Words: The Role of Externalization," *Linguistics and Philosophy* 5. <https://doi.org/10.1007/s10988-025-09447-7>
- Vicente, A. and J. Collins 2025. "The Expression of Thoughts: On Levelt's 'Message' and Thinking in Lexical Concepts," *Philosophical Psychology*. <https://doi.org/10.1080/09515089.2025.2609897>